SNAPDRAGON AI LAB | PROJECT DESCRIPTION

LectureLens

Offline lecture search with links back to the source

Problem

Students may know a concept was explained in a lecture but struggle to find the moment in a long recording or connect it to the slide being shown.

Solution

LectureLens puts the recording, timestamped transcript, and slide deck in one local workspace. Search returns source passages with an audio time or slide page, so a student can check the original explanation.

Working prototype

The Windows app imports audio, SRT/VTT/TXT transcripts, and PDF slides. It extracts selectable PDF text, renders slide pages, searches both sources locally, seeks to known audio times, and saves manual passage-to-slide links. The sample contains five transcript passages and three original slides.

Snapdragon-powered HP PC target

The app is intended for Windows 11 ARM64 on Snapdragon-powered HP PCs. An adapter calls Qualcomm's Whisper Windows sample with Qualcomm AI Hub model assets for offline WAV transcription. The adapter still requires target-device setup and QNN execution validation.

Demo and evidence

Run the app without Wi-Fi, import a lecture, search a concept, then open the cited passage and slide. Before making acceleration claims, verify the execution provider and measure transcription wall time on the same clip with CPU and QNN.

Source: Qualcomm AI Hub Whisper-Base and Qualcomm Whisper Windows sample.

Prototype status as of 28 September 2026. No NPU speed or battery results are claimed.